

design-research

Design & Technique Research — Cutting-Edge Portfolio + AI-Company Site Pair

Date: 2026-08-05 · **Scope:** milosvasic.ru (Jekyll, personal/crimson), vasic.digital (static, company/red), diagrams, downloadable PDFs (WeasyPrint). **Basis:** current (2025–2026) web research; every item has a concrete “how to apply here” + one source.

Hard constraints these must respect (see the “Constraint conflicts” section at the end for specifics): - **§11.4.162** — ALL design values live in **OpenDesign tokens consumed as a dependency**. No raw hex/px/rem literals and **no hand-authored CSS custom properties or inline styles** in site stylesheets. Any technique below that mentions `var(--*)`, `light-dark()`, `clamp()` type, etc. must be **authored in OpenDesign and emitted**, not hand-written. Missing capability → upstream PR (§11.4.74), never local CSS. - **Reduced-motion** — every motion technique gates on `@media (prefers-reduced-motion: no-preference)`. - **No-jank / GPU-only** — animate only transform, opacity, and compositor-safe filter; never width/height/top/left/box-shadow. - **Cross-browser** — scroll-driven animations and cross-document View Transitions are progressive-enhancement only (Chromium-led, Firefox/Safari lagging); the base experience must be complete without them. - **§11.4.170** — every surface proven by host-rendered pixels (golden image-diff + OCR/vision oracle) per screen × state × {light,dark}. Motion must be forced to a **deterministic end-state** for capture (see conflicts).

Priorities: **P0** = incorporate now / high leverage + low risk; **P1** = strong value, some effort or partial browser support; **P2** = nice-to-have / experimental.

1. High-end portfolio / AI-company website design + UX

- **P0 — Scrollytelling product narrative (structural, not decorative).** Tie hero → feature reveals to scroll position so each product capability enters as the reader arrives at it. *Apply:* on vasic.digital product pages, drive section reveals with CSS view() scroll timelines (Area 2) instead of a JS scroll library; keep content fully present without scroll for no-JS/reduced-motion. Source: <https://www.figma.com/resource-library/web-design-trends/>
 - **P0 — Typography-as-hero.** Oversized, confident headline + variable-font weight axis as the primary visual, reducing reliance on stock imagery. *Apply:* express the hero type scale and weight axis as OpenDesign typography tokens (self-hosted variable font per §11.4.97 plan); one bold H1 per page, tokenized. Source: <https://designmodo.com/web-design-trends/>
 - **P0 — Trust/credibility devices above the fold and per case study.** Logos, metrics (“X% faster”, repo/commit counts), named outcomes, and real screenshots build trust in an AI-saturated market. *Apply:* on both sites add a tokenized “proof strip” component (metrics + client/tech logos) and lead every portfolio detail page with a quantified result line; source metrics from the existing _analysis/github-*.json / gitlab-all.json inventories. Source: <https://www.utsubo.com/blog/web-design-trends-2026-decision-makers-guide>
 - **P1 — Case-study / detail-page pattern: Problem → Approach → Architecture diagram → Outcome → Artifacts.** A repeatable spine makes a portfolio read as senior/systematic. *Apply:* make this the fixed template for portfolio detail pages; the “Architecture diagram” slot reuses the site diagrams, the “Artifacts” slot links the downloadable PDF. Source: <https://www.theedigital.com/blog/web-design-trends>
 - **P1 — Human-crafted warmth signals (asymmetry, organic shape, custom illustration/texture).** Deliberate non-template craft differentiates from AI-generated sameness and reads as bespoke. *Apply:* introduce one signature asymmetric layout motif + custom diagram style shared across both brands (crimson vs red variants), tokenized so light/dark both hold. Source: <https://medium.com/creative-black-pug-studio/7-web-design-trends-being-shaped-by-ai-in-2026-4afbec683b34>
 - **P2 — Structural 3D tied to scroll/cursor (only if it earns its weight).** 3D as a storytelling layer, not a floating gimmick. *Apply:* reserve for a single company-page hero on vasic.digital; must degrade to a static tokenized image for reduced-motion/low-power and remain optional to the page. Source: <https://www.index.dev/blog/web-design-trends>
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2. Performant web animation & micro-interactions (zero-jank, GPU-only)

- **P0 — CSS scroll-driven animations (`animation-timeline: view() / scroll()`).** Bind animation progress to scroll on the compositor thread — no JS, no main-thread work, smooth at 120Hz even under load. *Apply:* replace any JS/IntersectionObserver reveal-on-scroll with `animation-timeline: view()` animating only opacity/transform; wrap in `@supports (animation-timeline: view()) and @media (prefers-reduced-motion: no-preference)` so non-supporting browsers (Firefox/Safari) and reduced-motion users get the static final state. Source: https://developer.mozilla.org/en-US/docs/Web/CSS/Guides/Scroll-driven_animations
- **P0 — `@starting-style` for enter animations.** Native “from” state for elements entering the DOM (modals, toasts, cards) — removes a JS/library dependency. *Apply:* animate the unified OpenDesign modal open and card mount with `@starting-style` on opacity/transform; pair with `transition-behavior: allow-discrete` for display. Source: <https://developer.chrome.com/blog/view-transitions-in-2025>
- **P0 — Reduced-motion as the default guard, not an afterthought.** Wrap motion in `prefers-reduced-motion: no-preference` and neutralize View Transitions with `::view-transition-group(*){animation-duration:.01s !important}` under `reduce`. *Apply:* add a single global reduced-motion partial to both sites’ OpenDesign motion layer; verify with a golden capture that forces `reduce`. Source: <https://developer.chrome.com/blog/view-transitions-in-2025>
- **P1 — Same-document View Transitions for in-page state (filters, tab/detail swaps).** Now reaching Baseline (Firefox 144, Oct 2025) for same-document; crossfade/shared-element morphs with almost no code. *Apply:* use `document.startViewTransition()` for portfolio filter/sort and light↔dark toggle on both sites; keep the default crossfade (don’t over-customize). Source: <https://developer.chrome.com/blog/view-transitions-in-2025>
- **P1 — Cross-document View Transitions + speculation rules for MPA navigation.** `@view-transition { navigation: auto; }` gives app-like page-to-page transitions on a static/Jekyll multi-page site; prerender via speculation rules. *Apply:* opt both the from- and to-page into `@view-transition` (both must opt in), gated on `prefers-reduced-motion: no-preference`; Chromium/Safari 18.2+ get it, others fall back to normal nav — no regression. Source: <https://css-tricks.com/cross-document-view-transitions-part-1/>
- **P1 — Prefer SVG/CSS over Lottie; if Lottie is unavoidable use `dotLottie`.** `lottie-web` is ~60–100 KB gzipped and runs on the main thread; SVG/CSS is 5–10× smaller for simple/medium motion. *Apply:* implement icon/micro-interactions as tokenized CSS/SVG (loading, hover, success); only reach for `.lottie`

(WASM, ~70–80% smaller container) for a genuinely complex designer-authored piece, lazy-loaded and reduced-motion-gated. Sources: <https://www.svgator.com/blog/svg-vs-lottie/> · <https://www.carmenansio.com/articles/animated-icon-libraries/>

- **P2 — Measure motion smoothness, don't eyeball it.** Track dropped-frame/jank ratio and keep animated props compositor-only. *Apply:* add a Motion-score/Lighthouse pass in CI and assert only transform/opacity/filter are animated (grep the emitted CSS in the §11.4.170 harness). Sources: <https://web.dev/articles/smoothness> · <https://score.motion.dev/>
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3. Design-system / design-token architecture

- **P0 — Three-tier tokens: primitive → semantic → component.** Primitives = raw palette; semantic = purpose (text/primary, surface/raised); component = per-component overrides. Theming changes only the semantic layer. *Apply:* structure the OpenDesign workspace in these three tiers so the crimson vs red brand swap and light/dark are single-layer remaps — matches the §11.4.162 “one foundation + two brand packages” decision. Source: <https://designsystemproblems.com/token-management/semantic-vs-primitive-tokens/>
- **P0 — DTCG token format + Style Dictionary v4 as the emit pipeline.** DTCG (2025.10) is the standard JSON contract; Style Dictionary v4 transforms it to per-platform output (CSS, SCSS, print). *Apply:* have OpenDesign emit DTCG, then Style Dictionary produce the three consumption targets the adoption plan needs — Jekyll SCSS, static-site CSS, and the WeasyPrint print stylesheet — from one source, no hand-authored vars. Source: <https://penpot.app/blog/the-developers-guide-to-design-tokens-and-css-variables/>
- **P1 — light-dark() for theming via a single token value.** One CSS value carries both modes: color: light-dark(<light>, <dark>), switched by color-scheme. *Apply:* have OpenDesign **emit** light-dark() into the semantic color tokens (do NOT hand-write it — §11.4.162), so one emitted stylesheet serves both themes and both golden captures. Source: <https://sujeet.pro/articles/design-tokens-and-theming>
- **P1 — Container queries for component-level responsiveness.** Components adapt to their container, not the viewport — essential for cards/diagrams reused in sidebars vs full-width. *Apply:* author the portfolio card, proof-strip, and diagram-caption components with @container so they reflow correctly wherever placed; express breakpoints as OpenDesign layout tokens. Source: <https://www.contentful.com/blog/design-token-system/>

- **P1 — Fluid type/space via clamp() emitted as tokens.** One responsive scale instead of stepped breakpoints. *Apply:* define the type/space scale in OpenDesign so it **emits** clamp() values into type tokens; sites consume var(--od-*) only. Source: <https://goodpractices.design/articles/design-tokens>
 - **P2 — Package tokens as a reusable, versioned library.** Semantic-layer stability lets you ship the system as a dependency others (or the PDF pipeline) consume. *Apply:* publish OpenDesign output as a versioned npm package feeding both sites + PDFs; treat it as the single source of truth per the adoption plan. Source: <https://designsystem.backbase.com/latest/design-tokens/introduction-5PSH8xS5>
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4. Enterprise-grade PDF / resume / portfolio (WeasyPrint)

- **P0 — Real print CSS with @page: margins, running headers/footers, page numbers.** WeasyPrint supports @page, margin boxes, @bottom-center { content: counter(page) }, and controlled breaks. *Apply:* build styles/default-pdf.css (A4, per §11.4.73) with a branded running footer (name · page X of Y) and a distinct first-page @page :first for the cover. Source: <https://print-css.rocks/>
- **P0 — Dedicated cover page with typographic hierarchy.** Full-bleed title, role, one-line value proposition, subtle brand mark; then break. *Apply:* Jinja2 template renders a cover section with the OpenDesign display type tokens + brand color, followed by break-after: page. Source: <https://www.incentius.com/blog-posts/build-modern-print-ready-pdfs-with-python-flask-weasyprint/>
- **P0 — Pagination hygiene: avoid orphaned headings / split blocks.** break-inside: avoid on cards/rows, break-after: avoid on headings, widows/orphans. *Apply:* apply to each résumé entry and portfolio card so no role/project splits across a page break — a top signal of a “top-notch” PDF. Source: <https://www.customjs.space/blog/print-css-cheatsheet/>
- **P1 — Compose PDF styling on OpenDesign tokens, not a separate palette.** One brand across web and print. *Apply:* feed the same emitted tokens into the print stylesheet (Style Dictionary print target) so the PDF’s color/type/spacing match the site exactly (§11.4.73 composition). Source: <https://weasyprint.org/>
- **P1 — Data-viz in PDFs via inline SVG (vector, print-crisp).** WeasyPrint renders embedded SVG/images cleanly; vector charts stay sharp at print DPI. *Apply:* embed the site’s architecture/skill diagrams as inline SVG styled by print tokens (not raster) so downloadable PDFs stay crisp and on-brand.

Source: <https://medium.com/aardvark-infinity/weasyprint-converting-html-and-css-to-high-quality-pdfs-50bdc3911514>

- **P2 — Self-host fonts + subset for the PDF.** Embedded, subsetted variable/weight family gives correct hierarchy and small file size. *Apply:* embed the same self-hosted family used on the sites (light/regular/bold/black) via `@font-face` in the print CSS. Source: <https://github.com/aiifabbbf/resume>
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5. Responsive / adaptive delivery per device

- **P0 — `<picture>` + `srcset/sizes` with AVIF→WebP→fallback.** AVIF ~50%+ smaller than JPEG (>90% support in 2026), WebP ~25-35% (~97%); saves 70-90% of bytes on phones. *Apply:* wire the already-existing (currently unused) WebP variants on milosvasic.ru into `<picture>`, add AVIF, and set correct sizes per layout slot. Source: <https://smolpix.co/blog/responsive-images-with-srcset-and-sizes-2026-guide>
 - **P0 — Container queries drive “the right version per component,” not just per viewport.** Same component, correct layout wherever placed. *Apply:* (as in Area 3) use `@container` so a diagram/card serves a compact vs full rendition by its own width. Source: <https://uploadcare.com/blog/responsive-images-vs-adaptive-delivery/>
 - **P0 — `loading="lazy"` + explicit width/height (CLS-safe).** Defer offscreen images; reserve space to avoid layout shift. *Apply:* lazy-load below-the-fold portfolio thumbnails and diagrams; always set intrinsic dimensions (also helps §11.4.170 stable goldens). Source: <https://www.xictron.com/en/blog/adaptive-image-loading-avif-lazy-shops-2026/>
 - **P1 — Honor data-saver / low-power intent.** `prefers-reduced-data` (and the Save-Data client hint) let you skip heavy hero media/animation for constrained users. *Apply:* gate the optional 3D/Lottie/large-hero behind `@media (prefers-reduced-data: no-preference)` and serve a tokenized static image otherwise; combine with the reduced-motion guard. Source: <https://uploadcare.com/blog/responsive-images-vs-adaptive-delivery/>
 - **P1 — Preload the LCP image, lazy everything else.** Explicitly prioritize the hero to protect LCP. *Apply:* `fetchpriority="high"` + preload the hero source per breakpoint; never lazy-load the LCP element. Source: <https://aibudwp.com/image-optimization-in-2025-webp-avif-srcset-and-preload/>
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6. Multilingual SEO + technical SEO (static/Jekyll, 15 languages)

- **P0 — Correct, fully-reciprocal hreflang clusters + x-default.** ~75% of implementations are broken; one bad/one-way tag makes Google ignore the whole cluster. *Apply:* generate hreflang in a Jekyll include from the language matrix so every one of the 15 locales links to all others **and back**, plus x-default; validate in CI (this is the single highest-risk SEO item for a 15-language site). Sources: <https://ulement.com/hreflang-tags-for-multilingual-sites/> · <https://434group.com/blog/hreflang-cok-dilli-seo.php?lang=en>
 - **P0 — JSON-LD entity graph: Organization + Person + WebSite, then per-page types.** Establish vasic.digital = Organization, Milos Vasic = Person (founder), each with stable @ids that cross-reference. *Apply:* one shared JSON-LD include; vasic.digital emits Organization, milosvasic.ru emits Person linked via sameAs/worksFor; product pages add SoftwareApplication, articles add TechArticle. Source: <https://squid.org/structured-data/organization-schema/>
 - **P0 — inLanguage on every schema + locale-aware structured data.** Declare each page's language explicitly in JSON-LD. *Apply:* inject the page's locale into inLanguage in the shared include so all 15 variants are unambiguous to search + AI crawlers. Source: <https://searchxpro.com/structured-data-for-multilingual-seo-top-7-tips/>
 - **P1 — SoftwareApplication schema on product pages.** Rich-result eligibility for the company's products. *Apply:* add SoftwareApplication (name, applicationCategory, operatingSystem, offers/aggregateRating only if real) to each vasic.digital product page. Source: https://dev.to/dheeraj_jain/a-developers-guide-to-implementing-json-ld-structured-data-for-better-technical-seo-nmg
 - **P1 — Localized sitemap with hreflang alternates.** A sitemap that enumerates every locale + its alternates is more robust than head tags alone at 15-language scale. *Apply:* generate sitemap.xml with xhtml:link rel="alternate" hreflang="..." per URL from the same language matrix. Source: <https://strapi.io/blog/multilingual-seo-best-practices>
 - **P1 — Core Web Vitals as a ranking gate (LCP/INP/CLS).** Only ~47% of sites pass; static/SSG + the image/motion discipline above gives near-instant LCP and low INP. *Apply:* the GPU-only motion (Area 2), preloaded LCP + lazy/srcset images (Area 5), and self-hosted fonts (no render-blocking Google Fonts) together target green CWV; assert in CI/Lighthouse. Source: <https://www.geniecrawl.com/the-ultimate-guide-to-core-web-vitals-optimization-in-2025/>
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Top 10 things to definitely incorporate

1. **Three-tier OpenDesign tokens (primitive→semantic→component), DTCG-emitted** feeding Jekyll SCSS + static CSS + WeasyPrint from one source — the backbone that keeps §11.4.162 satisfiable.
 2. **CSS scroll-driven animations (view()/scroll())** for all reveal-on-scroll — compositor-only, replaces JS scroll libs; behind @supports + reduced-motion.
 3. **Reciprocal hreflang clusters + x-default, CI-validated** across all 15 locales — the biggest technical-SEO risk, fixed once via a Jekyll include.
 4. **JSON-LD entity graph** (Organization ↔ Person ↔ WebSite, per-page SoftwareApplication/TechArticle, inLanguage everywhere).
 5. **<picture> + AVIF/WebP srcset/sizes + preload-LCP/lazy-rest** — wire up the existing unused WebP variants; 70–90% mobile byte savings and green LCP/CLS.
 6. **Trust/proof devices** (metrics strip, real screenshots, quantified outcomes) on hero + every portfolio detail page, sourced from the existing repo/GitLab inventories.
 7. **Repeatable case-study spine** (Problem→Approach→Architecture diagram→Outcome→PDF artifact) as the fixed portfolio detail template.
 8. **View Transitions** — same-document (now Baseline) for filters/theme toggle; cross-document + speculation rules for page nav; default crossfade, reduced-motion-gated.
 9. **Enterprise PDF craft** — @page cover + running footer/page numbers, break-inside: avoid pagination hygiene, inline-SVG data-viz, tokens shared with the sites.
 10. **@starting-style + GPU-only micro-interactions in CSS/SVG** (drop Lottie for simple motion; dotLottie only for genuinely complex pieces), honoring prefers-reduced-motion and prefers-reduced-data.
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Constraint conflicts to flag

- **light-dark(), clamp() fluid type, container-query breakpoints, @starting-style values — all must be EMITTED BY OpenDesign, never hand-authored (§11.4.162).** The techniques are endorsed, but the moment they appear as literal values or hand-written CSS custom properties in a site stylesheet they violate the no-escape-hatch mandate. If OpenDesign can't yet emit a construct (e.g. light-dark()), that's an **upstream PR** (§11.4.74), not a local var() definition.
- **Scroll-driven animations & cross-document View Transitions are Chromium-led; Firefox/Safari lag.** Must be strict progressive enhancement (@supports, both-page opt-in,

complete no-JS base). They cannot be the only way content becomes visible — otherwise the base render fails cross-browser and breaks the §11.4.170 golden on non-Chromium capture.

- **Motion vs §11.4.170 rendered-pixel proof (nondeterminism).** Scroll-driven/View-Transition intermediate frames are nondeterministic and will flap golden image-diffs. Capture must **force the deterministic end-state** — set prefers-reduced-motion: reduce (or seek animations to completion / disable animation-timeline) in the host-render harness so each screen × state × {light,dark} golden is stable. Add an explicit reduced-motion capture variant as its own state.
- **GPU-only rule bans common “trendy” props.** Any 3D/parallax/hover effect must animate only transform/opacity/compositor-safe filter; the CI check should grep emitted CSS and fail on animated width/height/top/left/box-shadow/margin.
- **AVIF/heavy hero media vs low-power/data-saver.** Gate optional 3D/Lottie/large hero behind prefers-reduced-data: no-preference **and** prefers-reduced-motion: no-preference, always with a tokenized static fallback — otherwise you regress CWV/INP on the exact low-end devices the “right version per device” goal targets.